THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 6: **We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Simplifying $7(x+3)-4(x-2)$ gives: 1

(a) $3x+29$ **(b)** $3x+13$ **(c)** $11x+29$ **(d)** $3x+16$

2. $(5x+3)(2x+4)$ equals: 1

(a) $10x^2+26x-12$ **(b)** $10x^2+26x+12$ **(c)** $10x^2-26x+12$ **(d)** $10x^2+12$

3. $(5x-2y)^2$ equals: 1

(a) $25x^2-4y^2$ **(b)** $25x^2-10xy+4y^2$ **(c)** $25x^2+20xy+4y^2$ **(d)** $25x^2-20xy+4y^2$

4. The value of 55^2-45^2 is: 1

(a) 5050 **(b)** 1000 **(c)** 100 **(d)** 10000

5. The value of $(x+2)^2$ when $x=9$ is: 1

(a) 85 **(b)** 125 **(c)** 103 **(d)** 121

6. The value of $(x+4)(x+7)$ when $x=2$ is: 1

(a) 54 **(b)** 56 **(c)** 32 **(d)** 26

7. $(x+7)^2-(x-7)^2$ equals: 1

(a) $2x^2+98$ **(b)** $14x$ **(c)** $28x$ **(d)** $28x+49$

8. Simplifying $4(x+6)-(x-3)$ gives: 1

(a) $5x+27$ **(b)** $3x+27$ **(c)** $3x+14$ **(d)** $3x+21$

9. $(3x+4)(4x+3)$ equals: 1

(a) $12x^2+25x-12$ **(b)** $12x^2+25x+12$ **(c)** $12x^2-25x+12$ **(d)** $12x^2+12$

10. $(3x-4y)^2$ equals: 1

(a) $9x^2-12xy+16y^2$ **(b)** $9x^2-16y^2$ **(c)** $9x^2-24xy+16y^2$ **(d)** $9x^2+24xy+16y^2$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Simplify $8(x+1)-3(x-4)$. 2

12. Expand $(5x+4y)^2$. 2

13. Find 82×98 using $(a+b)(a-b)$. 2

14. Simplify $6(x+4)-(x-5)$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Expand $(3x + 4)(4x - 3)$. 3

16. Find 98^2 using an algebraic identity. 3

17. Verify that $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$ for $a = 8$ and $b = 1$. 3

OR

Expand $(6x + 1)(2x - 3)$. 3

18. Find 203^2 using an algebraic identity. 3

OR

Verify that $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$ for $a = 10$ and $b = 7$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A square plot has side $(6x + 1)$ m. A square pond of side $4x$ m is dug inside it. Write the area of the remaining land as an expression in x and find it when $x = 10$. 5

OR

Use identities to evaluate: 5

(i) 99^2

(ii) 38×42

(iii) $57^2 - 43^2$.

20. A square plot has side $(5x + 4)$ m. A square pond of side $2x$ m is dug inside it. Write the area of the remaining land as an expression in x and find it when $x = 10$. 5

OR

Use identities to evaluate: 5

(i) 1004^2

(ii) 95×105

(iii) $72^2 - 28^2$.

END OF QUESTION PAPER — Total: 40 Marks

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